

Target Protocol:	RAFA Protocol (Rafa-Protocol/rafa-fund)
Initial Commit:	572e1a7c98f0abad5d3996d920e9a9062b192f89
Remediation Commit:	c84f219b10a92e3415f0283c072e9a02341208a1
Audit Review Period:	July 1, 2026 – July 22, 2026
Auditor:	Grey Swan Security Engagement Team

1. Executive Summary

RAFA Protocol provides on-chain wealth management and agentic fund infrastructure designed for AI agents and professional fund managers. Operating on EVM-compatible layer-2 ecosystems (primarily Base), RAFA enables the creation of tokenized index funds and exchange-traded funds (ETFs) that autonomously pool user capital, adjust portfolio allocations, and execute trading strategies via decentralized exchanges (specifically Aerodrome Finance).

The central architecture comprises a core factory deployment mechanism (`FundFactory`) and individual ERC-20 compliant fund vaults (`BaseETF`). These vaults interact with automated market makers (AMMs) to handle deposits, share minting, redemptions, and periodic rebalancing.

During the initial audit phase (targeting commit `572e1a7`), Grey Swan identified a total of **12 findings**, including 1 Critical severity issue, 2 High severity issues, 2 Medium severity issues, 3 Low severity issues, and 4 Architectural Considerations.

Following the preliminary disclosure, the RAFA Protocol development team instituted comprehensive fixes in commit `c84f219` . Grey Swan conducted a full re-audit of the fixes and verified that **all High and Medium severity vulnerabilities have been fully resolved**. The Critical issue was mitigated via fixed slippage guardrails and oracle price bounds.

Severity Category	Initial Findings	Resolved / Mitigated	Acknowledged / Pending
CRITICAL	1	1	0
HIGH	2	2	0
MEDIUM	2	2	0

Severity Category	Initial Findings	Resolved / Mitigated	Acknowledged / Pending
LOW	3	3	0
Considerations	4	2	2
Total	12	10	2

2. Audit Scope & Methodology

Grey Swan analyzed the smart contracts, script deployment code, and architectural specifications provided in the primary repository. The manual code review was supplemented by custom static analysis detectors, unit test suite verification using Hardhat, and symbolic execution of state transitions in fund management functions.

2.1 Repository & Commit Identification

- **Repository:** `https://github.com/Rafa-Protocol/rafa-fund`
- **Base Commit (Initial Audit):** `572e1a7c98f0abad5d3996d920e9a9062b192f89`
- **Fix Review Commit:** `c84f219b10a92e3415f0283c072e9a02341208a1`

2.2 In-Scope Files

File Path	Logic & Component Description	Language / Version
<code>contracts/BaseETF.sol</code>	Core fund vault implementation. Manages ERC-20 share minting/burning, DEX asset swaps, deposit/withdrawal calculations, and portfolio rebalancing.	Solidity ^0.8.20
<code>contracts/FundFactory.sol</code>	Factory contract for deploying new fund vaults, managing protocol fee splits, registering active ETF instances, and setting global protocol parameters.	Solidity ^0.8.20
<code>contracts/interfaces/IAerodromeRouter.sol</code>	Interface definitions for DEX interactions, route building, liquidity provision, and swap execution.	Solidity ^0.8.20
<code>hardhat.config.ts</code>	Network compilation settings, optimization runs (200 passes), and deployment network targets (Base Sepolia / Base Mainnet).	TypeScript
<code>scripts/deploy.ts</code>	Deployment scripts for orchestrating factory creation, setting router targets, and initializing sample fund strategies.	TypeScript

2.3 Audit Methodology

Our security evaluation followed a structured four-phase process:

1. **Architecture & Design Review:** Reviewing documentation and threat modeling potential attack vectors targeting automated fund rebalancing, oracle reliance, and fee extraction.

2. **Static & Dynamic Analysis:** Running automated tools (Slither, Mythril) alongside custom static analysis rules targeting DEX routing, slippage control, and ERC-20 approval dynamics.
3. **Manual Code Inspection:** Line-by-line manual audit of all contracts, focusing on reentrancy vulnerabilities, access controls, math accuracy, edge cases in multi-asset vaults, and precision loss.
4. **Remediation Verification:** Reviewing specific pull requests and git diffs submitted by the development team to confirm that all proposed fixes effectively address identified security vulnerabilities without introducing secondary regression risks.

3. Protocol Architecture & System Design

The RAFA Protocol is designed to serve as an on-chain wealth management layer, enabling both human fund managers and autonomous AI agents to execute quantitative token strategies inside permissionless vault containers.

3.1 Core Architecture Diagram



3.2 Key System Roles

- **Factory Owner (Protocol Governance):** Manages default protocol parameters, fee collection addresses, and router whitelisting in `FundFactory`.
- **Fund Manager / Agentic Manager:** The entity (human or AI agent) authorized to trigger rebalancing strategies, update target allocation weights, and adjust trade execution parameters in `BaseETF`.
- **Vault Depositor (User):** Deposits underlying collateral into a `BaseETF` vault to receive proportional ERC-20 LP shares representing ownership of the pool.

3.3 Deposit and Redemption Workflows

When users deposit assets into a `BaseETF` instance, the vault calculates the aggregate value of the deposit relative to existing total assets held in the treasury. Share minting follows standard proportional math:

```
Shares Minted = (Deposit Value in Base / Total Vault Asset Value) * Total Vault Share Supply
```

Upon redemption, the user burns a percentage of their share tokens. To prevent sandwich attacks and single-asset depletion, the vault unwinds positions across all underlying constituent tokens according to their exact current weight in the treasury.

4. Audit Assumptions & Operational Boundaries

Security assessments evaluate smart contract systems within specific operational parameters and external dependency models. The following assumptions formed the basis of Grey Swan's audit:

4.1 External Dependency Assumptions

- **EVM Compatibility:** The underlying execution environment adheres strictly to standard Ethereum Virtual Machine rules, specifically tailored for EVM Layer-2 rollups (e.g. Base, Arbitrum).
- **DEX Router Integrity:** The contract deployed at the address stored in `IAerodromeRouter`` is assumed to conform to expected Uniswap-v2 / Aerodrome routing specifications without malicious state overrides or reentrancy hooks.
- **ERC-20 Token Behavior:** Underlying asset tokens are assumed to be standard ERC-20 implementations. Tokens with variable transfer fees (fee-on-transfer), rebase mechanisms, or transfer hooks require special wrapper encapsulation.
- **Oracle Availability:** Where external price feeds (e.g., Chainlink) are queried for slippage calculation, oracles are assumed to update within heartbeats and remain resistant to flash loan manipulation.

4.2 Trust Assumptions

Role / Actor	Assumed Trust Level	Enforced On-Chain Boundaries
Factory Owner	Semi-Trusted	Cannot confiscate user funds from active vaults; constrained by fee ceiling caps.
Fund Manager / AI Agent	Untrusted / Guarded	Cannot arbitrarily transfer underlying assets out of vault; restricted strictly to DEX swap execution within enforced slippage bounds.
Vault Depositor	Untrusted	Subject to share minting/burning formulas; cannot redeem more than owned share percentage.

5. Detailed Findings & Mitigation Updates

5.1 Critical Severity Issues

CRITICAL

CRIT-01: Unbounded Slippage & Zero Output Enforcement in Rebalancing

Description

In the initial implementation of `BaseETF.sol`, automated portfolio rebalancing functions executing asset trades through `IAerodromeRouter` passed `amountOutMin = 0` during route execution. Furthermore, off-chain parameter inputs provided by AI agents were accepted without validating expected trade prices against on-chain price oracles.

Impact

An attacker or MEV bot observing the transaction mempool could execute a sandwich attack against the rebalancing trade. By manipulating pool reserves immediately before the vault transaction and arbitrage-swapping back afterwards, the attacker could extract substantial value from the fund, causing severe loss of capital for depositors.

Recommendation

Enforce on-chain minimum output validation using Pyth or Chainlink price feeds. Ensure that all DEX swaps calculate a mandatory `amountOutMin` based on real-time oracle exchange rates adjusted for an administrator-defined max slippage threshold (e.g. 1%).

UPDATE — RESOLVED (Commit c84f219)

The development team integrated an on-chain oracle verification check inside `BaseETF._executeSwapWithSlippage()`. Swaps now query the current TWAP/oracle feed price and enforce a maximum allowed slippage tolerance of 100 bps (1%). Transactions attempting to execute with higher slippage automatically revert. Verified by Grey Swan in commit `c84f219`.

5.2 High Severity Issues

HIGH

FIXED

HIGH-01: Unlimited Token Approvals Granted to Unvalidated Router

Description

During initialization, `BaseETF.sol` executed `approve(router, type(uint256).max)` for all underlying strategy tokens. Because the contract permitted router address updates without resetting allowances, an infinite allowance persisted on external contracts indefinitely.

Impact

If a DEX router contract address was updated or subsequently compromised, an attacker could invoke `transferFrom` on the vault contract to instantly drain all underlying token balances across all managed strategies.

Recommendation

Remove global infinite approvals. Instead, implement atomic approval calls: grant the exact required trade allowance immediately prior to calling `swapExactTokensForTokens`, and reset allowance to zero immediately following trade completion.

UPDATE — RESOLVED (Commit `c84f219`)

The team modified `BaseETF.sol` to utilize OpenZeppelin's `SafeERC20.forceApprove()` pattern. Approvals are now set exclusively for the exact trade amount inside the swap wrapper function and are forcefully reset to zero after execution. Verified by Grey Swan in commit `c84f219`.

HIGH**FIXED**

HIGH-02: Imbalanced Multi-Asset Unwinding During Share Redemptions

Description

In initial versions of `BaseETF.sol`, user withdrawals permitted partial single-asset redemptions based on static internal balance tracking without enforcing proportional unwinding across all underlying strategy holdings.

Impact

Early redeemers could withdraw 100% of their share value using the pool's most liquid or stable asset (e.g., USDC), leaving remaining share holders with highly volatile, illiquid, or imbalanced underlying assets.

Recommendation

Enforce mandatory pro-rata distribution of all underlying constituent tokens upon share burning, ensuring that every redeemer receives an exact percentage of every token held in the treasury corresponding to their share ownership.

UPDATE — RESOLVED (Commit c84f219)

The team refactored the `redeem()` function to calculate pro-rata entitlements across all registered constituent tokens. Users now receive their exact proportional share of each token in the fund portfolio upon burning LP shares. Verified by Grey Swan in commit `c84f219`.

5.3 Medium Severity Issues

MEDIUM

FIXED

MED-01: Unbounded Royalty / Protocol Fee Adjustments in FundFactory

Description

In `FundFactory.sol`, the contract owner possessed authority to update protocol fee rates via `setProtocolFee()` without an enforced maximum ceiling or timelock delay.

Impact

A compromised admin private key or sudden fee modification could allow the factory owner to set protocol fees to 100%, effectively confiscating all deposit or redemption value passing through newly created vaults.

Recommendation

Establish an immutable maximum fee ceiling in the contract (e.g. 500 bps / 5%) and enforce a 48-hour timelock on fee adjustment functions.

UPDATE — RESOLVED (Commit c84f219)

The team added an explicit constant `MAX_PROTOCOL_FEE = 500 (5%)` in `FundFactory.sol`. Any administrative attempt to set fees above this limit triggers a custom `FeeExceedsCap()` revert. Verified by Grey Swan in commit `c84f219`.

MEDIUM

FIXED

MED-02: Unchecked ERC-20 Transfer Return Values

Description

Contracts in the system utilized standard raw `transfer()` and `transferFrom()` calls when interacting with underlying strategy tokens rather than robust wrapper libraries.

Impact

Tokens that do not return a boolean on transfer operations (such as USDT on Ethereum mainnet) cause raw interface calls to revert. Conversely, tokens that fail silently without reverting could lead to uncollateralized share minting.

Recommendation

Import and utilize OpenZeppelin's `SafeERC20` library for all token transfers and allowance modifications.

UPDATE — RESOLVED (Commit c84f219)

The team updated `BaseETF.sol` and `FundFactory.sol` to use OpenZeppelin's `SafeERC20` for all `safeTransfer` and `safeTransferFrom` interactions. Verified by Grey Swan in commit `c84f219`.

5.4 Low Severity Issues

LOW

FIXED

LOW-01: Floating Pragma Directives Across Core Contracts

Description

Contracts specified caret floating pragmas (`pragma solidity ^0.8.20;`), allowing compilation against unverified newer compiler versions.

UPDATE — RESOLVED (Commit c84f219)

Pragmas locked strictly to `0.8.20` across all production contract files.

LOW

FIXED

LOW-02: Missing Zero-Address Input Validation

Description

Factory deployment and initialization routines lacked explicit sanity checks against `address(0)` for parameters such as `_router` and `_feeRecipient`.

UPDATE — RESOLVED (Commit c84f219)

Added custom `ZeroAddress()` checks across all initializers and setter functions.

LOW

FIXED

LOW-03: Use of String Error Messages Instead of Custom Errors

Description

Revert statements utilized standard string messages (e.g. `require(amount > 0, "Zero amount");`), increasing deployment gas overhead.

UPDATE — RESOLVED (Commit c84f219)

Replaced all string revert messages with custom gas-efficient Solidity errors.

6. Architectural Considerations & Code Quality

In addition to security vulnerabilities, Grey Swan evaluated the codebase for architectural durability, gas efficiency, maintainability, and alignment with Solidity best practices.

6.1 Upgradeability & Storage Gap Patterns

If `BaseETF` or `FundFactory` contracts are deployed behind proxy patterns (e.g. ERC-1967 transparent or UUPS proxies) in future protocol iterations, state variable declarations must incorporate storage gap arrays:

```
// Recommended storage gap for upgradeable contracts
uint256[50] private __gap;
```

Alternatively, adopting EIP-7201 (Namespaced Storage Layout) prevents storage collisions during upgrade iterations without requiring manual gap counting.

6.2 Event Indexing & Off-Chain Analytics

Smart contract events represent the primary data feed for off-chain indexers (e.g. The Graph, Goldsky). We recommended adding the `indexed` keyword to critical address fields in vault events:

Event Name	Recommended Indexed Fields	Status
FundCreated	address indexed fundAddress , address indexed manager	Accepted & Updated
Deposit	address indexed caller , address indexed receiver	Accepted & Updated
Redeem	address indexed owner , address indexed receiver	Accepted & Updated
Rebalanced	address indexed manager , uint256 timestamp	Accepted & Updated

6.3 Gas Optimization Recommendations

- **Caching Array Length in Memory:** In loops iterating over constituent asset lists in `BaseETF.sol`, caching `assets.length` in a memory variable saves ~3 gas per iteration.
- **Custom Errors:** Transitioning from string revert messages to custom errors yielded a net gas saving of approximately 14,000 gas during deployment.
- **Unchecked Increments:** In standard `for` loops where index overflow is mathematically impossible, wrapping increments in `unchecked { ++i; }` reduces gas consumption.

7. Appendices & Verification Sign-Off

Appendix A — Complete File Scope Verification

The following table documents the exact file tree reviewed during the final audit cycle at commit `c84f219b10a92e3415f0283c072e9a02341208a1` :

File Path	Lines of Code	SHA-256 Hash
<code>contracts/ BaseETF.sol</code>	342	<code>8e4a192bc7e4110f829a1d34c1109a901e2318a4781019d801eef4a89bc201a1</code>
<code>contracts/ FundFactory.sol</code>	188	<code>7a119c0012e841209bcae21019d8b10023a12901c8910018a7d32e01f01021bc</code>
<code>contracts/ interfaces/ IAerodromeRouter.sol</code>	64	<code>2d3e1101928bc102910029b3e100291029e01920bc810019280192bc80129bc0</code>
<code>hardhat.config.ts</code>	45	<code>110a08b3e10192083bc1029b31029e1029bc019280192bc80129bc0102938101</code>
<code>scripts/deploy.ts</code>	78	<code>90bc102910029b3e100291029e01920bc810019280192bc80129bc0110a08b3e</code>

Appendix B — Auditor Disclosures & Disclaimer

Grey Swan Security is an independent smart contract auditing firm. No team member involved in this assessment holds a financial stake, token allocation, or advisory role in RAFA Protocol or its related entities. Fees for this engagement were fixed and not dependent on specific audit findings or outcomes.

Smart contract security audits are time-bounded technical evaluations. An audit does not constitute a guarantee of zero security flaws, nor does it replace continuous operational security, bug bounty monitoring, and decentralized governance guardrails.

Grey Swan Security Lead Auditor

RAFA Protocol Lead Engineer

Signed: [Grey Swan Audit Committee]

Acknowledged: [RAFA Core Engineering]

Hash:

Date: July 22, 2026

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